

```
thislist=["apple","banana","cherry"]
print(thislist)

['apple', 'banana', 'cherry']

thislist=["apple","banana","cherry","apple","cherry"]
print(thislist)

['apple', 'banana', 'cherry', 'apple', 'cherry']

thislist=["apple","banana","cherry"]
print(len(thislist))

3

list1=["apple","banana","cherry"]
list2=[1,5,7,9,3]
list3=[True,False,False]

list1=["abc",34,True,40,"male"]

mylist=["apple","banana","cherry"]
print(type(mylist))

<class 'list'>

thislist=list(("apple","banana","cherry"))
print(thislist)

['apple', 'banana', 'cherry']

thislist=["apple","banana","cherry"]
print(thislist[1])

banana

thislist=["apple","banana","cherry"]
print(thislist[-1])

cherry

thislist=["apple","banana","cherry","orange","kiwi","melon","mango"]
print(thislist[2:5])

['cherry', 'orange', 'kiwi']

thislist=["apple","banana","cherry","orange","kiwi","melon","mango"]
print(thislist[:4])

['apple', 'banana', 'cherry', 'orange']

thislist=["apple","banana","cherry","orange","kiwi","melon","mango"]
print(thislist[2:])
```

```

['cherry', 'orange', 'kiwi', 'melon', 'mango']

thislist=["apple","banana","cherry","orange","kiwi","melon","mango"]
print(thislist[-4:-1])

['orange', 'kiwi', 'melon']

thislist=["apple","banana","cherry"]
if "apple" in thislist:
    print("yes,'apple' is in the fruits list")

yes,'apple' is in the fruits list

thislist=["apple","banana","cherry"]
thislist[1]="blackcurrant"
print(thislist)

['apple', 'blackcurrant', 'cherry']

thislist=["apple","banana","cherry","orange","kiwi","mango"]
thislist[1:3]=["blackcurrent","watermelon"]
print(thislist)

['apple', 'blackcurrent', 'watermelon', 'orange', 'kiwi', 'mango']

thislist=["apple","banana","cherry"]
thislist[1:2]=["balckcurrant","watermelon"]
print(thislist)

['apple', 'balckcurrant', 'watermelon', 'cherry']

thislist=["apple","banana","cherry"]
thislist[1:3]=["watermelon"]
print(thislist)

['apple', 'watermelon']

thislist=["apple","banana","cherry"]
thislist.insert(2,"watermelon")
print(thislist)

['apple', 'banana', 'watermelon', 'cherry']

thislist=["apple","banana","cherry"]
thislist.append("orange")
print(thislist)

['apple', 'banana', 'cherry', 'orange']

thislist=["apple","banana","cherry"]
thislist.insert(1,"orange")
print(thislist)

```

```
['apple', 'orange', 'banana', 'cherry']

thislist=["apple","banana","cherry"]
tropical=["mango","pineapple","papaya"]
thislist.extend(tropical)
print(thislist)

['apple', 'banana', 'cherry', 'mango', 'pineapple', 'papaya']

thislist=["apple","banana","cherry"]
thistuple=("kiwi","orange")
thislist.extend(thistuple)
print(thislist)

['apple', 'banana', 'cherry', 'kiwi', 'orange']

thislist=["apple","banana","cherry"]
thislist.remove("banana")
print(thislist)

['apple', 'cherry']

thislist=["apple","banana","cherry","banana","kiwi"]
thislist.remove("banana")
print(thislist)

['apple', 'cherry', 'banana', 'kiwi']

thislist=["apple","banana","cherry"]
thislist.pop(1)
print(thislist)

['apple', 'cherry']

thislist=["apple","banana","cherry"]
thislist.pop()
print(thislist)

['apple', 'banana']

thislist=["apple","banana","cherry"]
del thislist[0]
print(thislist)

['banana', 'cherry']

thislist=["apple","banana","cherry"]
del thislist

thislist=["apple","banana","cherry"]
thislist.clear()
print(thislist)
```

```
[]
```

```
thislist=["apple","banana","cherry"]  
for x in thislist:  
    print(x)
```

```
apple  
banana  
cherry
```

```
thislist=["apple","banana","cherry"]  
for i in range(len(thislist)):  
    print(thislist[i])
```

```
apple  
banana  
cherry
```

```
thislist=["apple","banana","cherry"]  
i=0  
while i < len(thislist):  
    print(thislist[i])  
    i=i+1
```

```
apple  
banana  
cherry
```

```
fruits=["apple","banana","cherry","kiwi","mango"]  
newlist=[]
```

```
for x in fruits:  
    if "a" in x:  
        newlist.append(x)
```

```
print(newlist)
```

```
['apple', 'banana', 'mango']
```

```
fruits=["apple","banana","cherry","kiwi","mango"]  
newlist=[x for x in fruits if "a" in x]  
print(newlist)
```

```
['apple', 'banana', 'mango']
```

```
fruits=["apple","banana","cherry","kiwi","mango"]  
newlist=[x for x in fruits if x != "apple"]  
print(newlist)
```

```
newlist=[x for x in fruits]  
print(newlist)
```

```

newlist=[x for x in range(10)]
print(newlist)

newlist=[x for x in range(10) if x<5]
print(newlist)

newlist=[x.upper() for x in fruits]
print(newlist)

newlist=['hello' for x in fruits]
print(newlist)

newlist=[x if x != "banana" else "orange" for x in fruits]
print(newlist)

['banana', 'cherry', 'kiwi', 'mango']
['apple', 'banana', 'cherry', 'kiwi', 'mango']
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
[0, 1, 2, 3, 4]
['APPLE', 'BANANA', 'CHERRY', 'KIWI', 'MANGO']
['hello', 'hello', 'hello', 'hello', 'hello']
['apple', 'orange', 'cherry', 'kiwi', 'mango']

thislist=["orange","mango","kiwi","pineapple","banana"]
thislist.sort()
print(thislist)

['banana', 'kiwi', 'mango', 'orange', 'pineapple']

thislist=[100,50.65,82,23]
thislist.sort()
print(thislist)

[23, 50.65, 82, 100]

thislist=["orange","mango","kiwi","pineapple","banana"]
thislist.sort(reverse=True)
print(thislist)

['pineapple', 'orange', 'mango', 'kiwi', 'banana']

thislist=[100,50.65,82,23]
thislist.sort(reverse=True)
print(thislist)

[100, 82, 50.65, 23]

def myfunc(n):
    return abs(n-50)

thislist=[100,50,65,82,23]

```

```
thislist.sort(key=myfunc)
print(thislist)

[50, 65, 23, 82, 100]

thislist=["banana","orange","kiwi","cherry"]
thislist.sort()
print(thislist)

['banana', 'cherry', 'kiwi', 'orange']

thislist=["banana","orange","kiwi","cherry"]
thislist.sort(key=str.lower)
print(thislist)

['banana', 'cherry', 'kiwi', 'orange']

thislist=["banana","orange","kiwi","cherry"]
thislist.reverse()
print(thislist)

['cherry', 'kiwi', 'orange', 'banana']

thislist=["apple","banana","cherry"]
mylist=thislist[:]
print(mylist)

['apple', 'banana', 'cherry']

list1=["a","b","c"]
list2=[1,2,3]

list3=list1+list2
print(list3)

['a', 'b', 'c', 1, 2, 3]

list1=["a","b","c"]
list2=[1,2,3]

list1.extend(list2)
print(list1)

['a', 'b', 'c', 1, 2, 3]

thistuple=("apple","banana","cherry")
print(thistuple)

('apple', 'banana', 'cherry')

thistuple=("apple","banana","cherry","apple","cherry")
print(thistuple)
```

```

('apple', 'banana', 'cherry', 'apple', 'cherry')
thistuple=("apple","banana","cherry")
print(len(thistuple))
3
thistuple=("apple",)
print(type(thistuple))
thistuple=("apple")
print(type(thistuple))
<class 'tuple'>
<class 'str'>
tuple1=("apple","banana","cherry")
tuple2=(1,5,7,9,3)
tuple3=(True,False,False)
mytuple=("apple","banana","cherry")
print(type(mytuple))
<class 'tuple'>
thistuple=tuple(("apple","banana","cherry"))
print(thistuple)
('apple', 'banana', 'cherry')
thistuple=("apple","banana","cherry")
print(thistuple[1])
banana
thistuple=("apple","banana","cherry")
print(thistuple[-1])
cherry
thistuple=("apple","banana","cherry","orange","kiwi","melon","mango")
print(thistuple[2:5])
('cherry', 'orange', 'kiwi')
thistuple=("apple","banana","cherry","orange","kiwi","melon","mango")
print(thistuple[:4])
('apple', 'banana', 'cherry', 'orange')
thistuple=("apple","banana","cherry","orange","kiwi","melon","mango")
print(thistuple[2:])
('cherry', 'orange', 'kiwi', 'melon', 'mango')

```

```

thistuple=("apple","banana","cherry","orange","kiwi","melon","mango")
print(thistuple[-4:-1])

('orange', 'kiwi', 'melon')

thistuple=("apple","banana","cherry")
if "apple" in thistuple:
    print("yes,'apple' is in the fruits tuple")

yes,'apple' is in the fruits tuple

x=("apple","banana","cherry")
y=list(x)
y[1]="kiwi"
x=tuple(y)

print(x)

('apple', 'kiwi', 'cherry')

thistuple=("apple","banana","cherry")
y=list(thistuple)
y.append("orange",)
thistuple=tuple(y)
print(thistuple)

('apple', 'banana', 'cherry', 'orange')

thistuple=("apple","banana","cherry")
y=("orange",)
thistuple += y

print(thistuple)

('apple', 'banana', 'cherry', 'orange')

thistuple=("apple","banana","cherry")
y=list(thistuple)
y.remove("apple")
thistuple=tuple(y)

print(thistuple)

('banana', 'cherry')

thistuple=("apple","banana","cherry")
del thistuple

fruit=("apple","banana","cherry")
print(fruits)

['apple', 'banana', 'cherry', 'kiwi', 'mango']

```

```
fruits=("apple","banana","cherry")

(green,yellow,red)=fruits

print(green)
print(yellow)
print(red)

apple
banana
cherry

fruits=("apple","banana","cherry","strawberry","raspberry")

(green,yellow,*red)=fruits

print(green)
print(yellow)
print(red)

apple
banana
['cherry', 'strawberry', 'raspberry']

fruits=("apple","mango","pineapple","cherry")

(green,*tropic,red)=fruits

print(green)
print(yellow)
print(red)

apple
banana
cherry

thistuple=("apple","banana","cherry")
for x in thistuple:
    print(x)

apple
banana
cherry

thistuple=("apple","banana","cherry")
for i in range(len(thistuple)):
    print(thistuple[i])

apple
banana
cherry
```

```
thistuple=("apple","banana","cherry")
i=0
while i < len(thistuple):
    print(thistuple[i])
    i=i+1
```

```
apple
banana
cherry
```

```
tuple1=("a","b","c")
tuple2=(1,2,3)
```

```
tuple3=tuple1+tuple2
print(tuple3)
```

```
('a', 'b', 'c', 1, 2, 3)
```

```
fruits=("apple","banana","cherry")
mytuple=fruits*2
```

```
print(mytuple)
```

```
('apple', 'banana', 'cherry', 'apple', 'banana', 'cherry')
```